Day - 6

Block 1 - Aptitude

permutation & combination.

1) 3 students from 7. $\frac{7\times6\times5\times4\times3\times2\times1}{3\times2\times1\ \ 4\times3\times2\times1} = 7\times5 = 35$

$$^7C_3 = \frac{7!}{3!\,4!} = \frac{5040}{24} = 35 \text{ ways}$$

2) First digit = 4
second digit = 3

Total = $4\times3 = 12$

3) Captain & vice captain from 6 players

$$^6P_2 = \frac{6!}{4!} = \frac{6\times5\times4\times3\times2\times1}{4\times3\times2\times1}$$

$= 30$ way

4) MATH (unique)

$4! = 4\times3\times2\times1$

$= 24$ arrange

5) $$^8C_4 = \frac{8\times7\times6\times5\times4\times3\times2\times1}{(4\times3\times2\times1)(4\times3\times2\times1)}$$

$= 70$ way

Block 2 - Verbal ability

Reading passage.

1) Databases are important because they help in efficient storage, retrieval and managing the data.

2) Data consistency means maintaining accurate and uniform data across the system.

3) Companies need them to ensure better performance and accurate data handling.

Grammar correction

1) She do not understand SQL joins.
She does not understand SQL joins

2) He were explaining the database schema
He was explaining the database schema

3) They has written the program successfully
They have written the program successfully

Block 3 - Communication

One challenge I faced while learning coding was converting logic into code.

Whenever I solved a problem I usually understood the logic and approach, but I struggled to implement it in code

To overcome this, I started practicing more problems and focused on writing step-by-step solutions.

I also broke the logic into smaller parts and then converted each step into code.

Over time, this improved my confidence.

From this, I learned that coding is not just about understanding logic, but also about practicing and implementing it consistently.